

KalaSetu Mobile Application — Log Messages Study Sheet

This document catalogs every `debugPrint` log message instrumented across the 27 presentation screens of the **KalaSetu** Flutter mobile application. It is designed for developer onboarding, QA verification, debugging, and production telemetry monitoring.

1. Logging Strategy & Tagging Architecture

Every log statement follows a structured, searchable convention:

```
[ScreenClassName] ActionOrEvent: Contextual parameters
```

Filtering in Terminal or IDE

Filter by Screen: `debugPrint('[ArtisanCameraScreen]')`

Filter by Flow: `debugPrint('[LoginScreen]')`, `debugPrint('[ShopCatalogScreen]')`

Via Terminal (ADB/macOS/Linux/Windows PowerShell):

```
flutter logs | Select-String "[Artisan"
```

```
adb logcat -s flutter | grep "\["
```

2. Master Log Messages Reference Table

Group 1: Auth, Roles & Onboarding (7 Screens)

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
1	SplashScreen	features/auth/presentation/splash_screen.dart	initState / App Mount	[SplashScreen] Initializing splash screen ...	Tracks initial launch and cold start sequence.
1	SplashScreen	features/auth/presentation/splash_screen.dart	Auth Check (Authenticated)	[SplashScreen] Authenticated user found (role: \$role) → routing to \$destination	Validates session persistence and role-based redirect.
1	SplashScreen	features/auth/presentation/splash_screen.dart	Auth Check (Unauthenticated)	[SplashScreen] No session found → navigating to /language	Confirms onboarding kickoff for first-time / logged-out users.
2	LanguageSelectScreen	features/auth/presentation/language_select_screen.dart	build render	[LanguageSelectScreen] Rendered. Selected: \$selectedLocale	Confirms locale detection on boot.
2	LanguageSelectScreen	features/auth/presentation/language_select_screen.dart	Language Option Tap	[LanguageSelectScreen] Selected language: Hindi / English	Captures language preference toggle.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
2	LanguageSelectScreen	features/auth/presentation/language_select_screen.dart	Continue Button	[LanguageSelectScreen] Language saved (\$selectedLocale) → navigating to /role-select	Confirms locale state commit before role selection.
3	RoleSelectScreen	features/auth/presentation/role_select_screen.dart	build render	[RoleSelectScreen] Rendered	Tracks role selection screen presentation.
3	RoleSelectScreen	features/auth/presentation/role_select_screen.dart	Role Card Tap	[RoleSelectScreen] Selected role: Artisan / Customer	Captures user persona selection before login.
3	RoleSelectScreen	features/auth/presentation/role_select_screen.dart	Role Confirmation	[RoleSelectScreen] Navigating to role destination: /login or /buyer/login	Tracks bifurcation into Artisan or Buyer authentication flow.
4	LoginScreen (Artisan)	features/auth/presentation/login_screen.dart	Phone Validation Error	[LoginScreen] Phone validation error: Phone number must be 10 digits	Monitors validation drop-off points.
4	LoginScreen (Artisan)	features/auth/presentation/login_screen.dart	Send OTP Attempt	[LoginScreen] Attempting OTP send for phone: \$phone	Diagnoses authentication network requests.
4	LoginScreen (Artisan)	features/auth/presentation/login_screen.dart	OTP Sent Success	[LoginScreen] OTP sent successfully → navigating to /otp-verify	Confirms SMS gateway handshake.
4	LoginScreen (Artisan)	features/auth/presentation/login_screen.dart	OTP Failure	[LoginScreen] Failed to send OTP: \$error	Error tracking for gateway failures.
4	LoginScreen (Artisan)	features/auth/presentation/login_screen.dart	Google Sign-In	[LoginScreen] Google Sign-In tapped	Monitors third-party OAuth entry.
5	BuyerLoginScreen	features/auth/presentation/buyer_login_screen.dart	Send OTP Attempt	[BuyerLoginScreen] Attempting OTP send for phone: \$phone	Tracks buyer-specific authentication traffic.
5	BuyerLoginScreen	features/auth/presentation/buyer_login_screen.dart	OTP Sent Success	[BuyerLoginScreen] OTP sent successfully → navigating to /otp-verify	Tracks buyer OTP routing.
5	BuyerLoginScreen	features/auth/presentation/buyer_login_screen.dart	Google Sign-In	[BuyerLoginScreen] Google Sign-In tapped	Tracks buyer OAuth login.
6	SignupScreen	features/auth/presentation/signup_screen.dart	Screen Render	[SignupScreen] Rendered	Monitors artisan self-registration funnel.
6	SignupScreen	features/auth/presentation/signup_screen.dart	Form Submission	[SignupScreen] Submitting registration for name: \$name, phone: \$phone, craft: \$craft	Audits artisan registration inputs and craft taxonomy.
7	OtpVerifyScreen	features/auth/presentation/otp_verify_screen.dart	initState	[OtpVerifyScreen] Initialized. Phone: \$phone, destination: \$destination	Validates target parameters passed to verification view.
7	OtpVerifyScreen	features/auth/presentation/otp_verify_screen.dart	Verify Button	[OtpVerifyScreen] Verifying code: \$code for phone: \$phone	Tracks verification attempts.
7	OtpVerifyScreen	features/auth/presentation/otp_verify_screen.dart	Verification Success	[OtpVerifyScreen] OTP verified successfully → routing to \$destination	Confirms authentication grant and dashboard routing.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
7	OtpVerifyScreen	features/auth/presentation/otp_verify_screen.dart	Verification Failure	[OtpVerifyScreen] OTP verification failed: \$error	Captures wrong code or timeout errors.
7	OtpVerifyScreen	features/auth/presentation/otp_verify_screen.dart	Resend OTP	[OtpVerifyScreen] Resending OTP to \$phone	Telemetry for OTP resend fatigue and SMS retry logic.

Group 2: Artisan Studio & Add Item Wizard (8 Screens)

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	initState	[ArtisanHomeScreen] Initialized	Verifies artisan home dashboard mount.
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	Audio Guide Toggle	[ArtisanHomeScreen] Audio guide playback toggled: isPlaying=\$isPlaying	Tracks voice assistance usage among artisans.
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	Camera Hero Card	[ArtisanHomeScreen] Camera hero card tapped → /artisan/add-item/camera	Funnel tracking: Start of product creation.
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	Metric Tile Click	[ArtisanHomeScreen] Metric tile tapped: earnings / orders / fairs	Telemetry for dashboard tile interest.
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	View All Crafts	[ArtisanHomeScreen] View All Crafts tapped → /artisan/shop-catalog/art_1	Verifies catalog navigation fix.
8	ArtisanHomeScreen	features/artisan_home/presentation/artisan_home_screen.dart	Product Card Tap	[ArtisanHomeScreen] Product tapped: id=\$id, nameEn="\$name"	Tracks specific item management clicks.
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	initState	[ShopCatalogScreen] Initialized for artisanId: \$artisanId	Validates shop ID mapping and catalog data loading.
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	Search Query Changed	[ShopCatalogScreen] Search query changed: "\$query"	Monitors live in-catalog search queries.
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	Search Cleared	[ShopCatalogScreen] Search query cleared	Search reset interaction tracking.
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	Category Chip	[ShopCatalogScreen] Category filter selected: \$category	Audits craft category filtering behavior.
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	Product Card Tap	[ShopCatalogScreen] Product card tapped: id=\$id, nameEn="\$name", isBuyer=\$isBuyer	Verifies contextual routing between artisan and buyer views.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
9	ShopCatalogScreen	features/artisan_home/presentation/shop_catalog_screen.dart	Add Craft FAB	[ShopCatalogScreen] Add Craft FAB pressed → navigating to /artisan/add-item/camera	Funnel tracking: New craft creation entry.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	initState / Permissions	[ArtisanCameraScreen] Requesting camera permissions... Granted: \$granted	Diagnoses device hardware camera permissions.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Camera Initialized	[ArtisanCameraScreen] Camera initialized. Ready: \$isInitialized	Verifies live camera preview controller readiness.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Shutter Button Tap	[ArtisanCameraScreen] Shutter pressed. Flash state: \$flashState	Audits camera shutter latency and flash configuration.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Photo Capture Result	[ArtisanCameraScreen] Photo captured: \$photoPath, navigating to photo review	Captures saved image file URI and wizard state transition.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Gallery Pick	[ArtisanCameraScreen] Gallery photo selected: \$path	Tracks gallery image importation.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Flash Toggle	[ArtisanCameraScreen] Flash toggled: state=\$state, mode=\$mode	Audits flash cycle (off, on, auto).
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Torch Toggle	[ArtisanCameraScreen] Toggling torch... Torch is now: \$isTorchOn	Verifies torch light state.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Flip Camera	[ArtisanCameraScreen] Switch camera requested. Result: \$switched	Verifies front/back camera lens switching.
10	ArtisanCameraScreen	features/artisan_add_item/presentation/artisan_camera_screen.dart	Voice Note Toggle	[ArtisanCameraScreen] Voice note toggled: isRecording=\$isVoiceRecording	Quick audio description toggle on camera viewfinder.
11	ImageCropperScreen	features/artisan_add_item/presentation/image_cropper_screen.dart	initState	[ImageCropperScreen] Initialized with image: \$imagePath	Validates target image passed to cropper.
11	ImageCropperScreen	features/artisan_add_item/presentation/image_cropper_screen.dart	Aspect Ratio Select	[ImageCropperScreen] Aspect ratio selected: \$ratio	Logs preset ratio selection (1:1, 4:5, 16:9, free).
11	ImageCropperScreen	features/artisan_add_item/presentation/image_cropper_screen.dart	Reset Crop	[ImageCropperScreen] Reset crop parameters	Tracks user resetting cropping box.
11	ImageCropperScreen	features/artisan_add_item/presentation/image_cropper_screen.dart	Rotate Clockwise	[ImageCropperScreen] Rotate 90 deg clockwise. New turns: \$turns	Tracks 90-degree image rotation steps.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
11	ImageCropperScreen	features/artisan_add_item/presentation/image_cropper_screen.dart	Crop Confirmed	[ImageCropperScreen] Crop confirmed: rect=(\$l, \$t, \$r, \$b), rotation=\$turns	Logs final bounding crop box coordinates.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	initState	[PhotoReviewScreen] Initialized. photoPath: \$path	Verifies preview stage mount with captured photo.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	Aspect Ratio Resolved	[PhotoReviewScreen] Aspect ratio resolved: \$ratio (dimensions: \${w}x\${h})	Prevents yellow distortion by logging dimensions.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	Before/After Toggle	[PhotoReviewScreen] Switched to AI Studio Backdrop / Original Photo view	Tracks artisan comparison of AI background treatment.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	Tap to Crop	[PhotoReviewScreen] Tapped photo to crop → opening /artisan/add-item/crop	Cropping workflow transition.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	Retake Button	[PhotoReviewScreen] Retake button tapped → popping to camera	Audits photo retake rate.
12	PhotoReviewScreen	features/artisan_add_item/presentation/photo_review_screen.dart	Next Button	[PhotoReviewScreen] Next button tapped → navigating to voice describe	Step completion tracking.
13	VoiceDescribeScreen	features/artisan_add_item/presentation/voice_describe_screen.dart	initState	[VoiceDescribeScreen] Initialized. Starting recording setup ...	Mount audit for microphone recorder.
13	VoiceDescribeScreen	features/artisan_add_item/presentation/voice_describe_screen.dart	Record Start	[VoiceDescribeScreen] Audio recording active at path: \$filePath	Logs temporary audio file creation.
13	VoiceDescribeScreen	features/artisan_add_item/presentation/voice_describe_screen.dart	Pause / Resume	[VoiceDescribeScreen] Toggle pause. Current isPaused: \$isPaused	Tracks audio pause/resume states.
13	VoiceDescribeScreen	features/artisan_add_item/presentation/voice_describe_screen.dart	Re-record Button	[VoiceDescribeScreen] Re-record button tapped	Tracks audio retake rate.
13	VoiceDescribeScreen	features/artisan_add_item/presentation/voice_describe_screen.dart	Finish Recording	[VoiceDescribeScreen] Recorded audio saved: \$path. Navigating to AI review	Saves audio note to wizard and navigates to AI analysis.
14	AiReviewScreen	features/artisan_add_item/presentation/ai_review_screen.dart	initState	[AiReviewScreen] Initialized. Prepopulated AI title & description ready for review.	Audits AI generation arrival.
14	AiReviewScreen	features/artisan_add_item/presentation/ai_review_screen.dart	Set Pricing Tap	[AiReviewScreen] Set Pricing tapped. nameHi: "\$nameHi", nameEn: "\$nameEn"	Captures artisan overrides of AI-generated titles.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
15	PricingScreen	features/artisan_add_item/presentation/pricing_screen.dart	initState	[PricingScreen] Initialized. suggestedPrice=\$price, rawMaterial=\$mat, laborHours=\$hrs	Audits fair price cost calculation.
15	PricingScreen	features/artisan_add_item/presentation/pricing_screen.dart	Stock Counter +/-	[PricingScreen] Stock increased / decreased to: \$stock	Tracks inventory stock adjustments.
15	PricingScreen	features/artisan_add_item/presentation/pricing_screen.dart	Publish Item	[PricingScreen] Publishing item with price: ₹\$price, stock: \$stock	Captures final listing commit.
15	PricingScreen	features/artisan_add_item/presentation/pricing_screen.dart	Publish Success Dialog	[PricingScreen] Item published successfully. Showing confirmation modal.	Confirms marketplace listing publication.
15	PricingScreen	features/artisan_add_item/presentation/pricing_screen.dart	Back to Home	[PricingScreen] Back to home tapped → resetting wizard and navigating to /artisan/home	Confirms wizard cleanup and return to studio.

Group 3: Artisan Management & Profile (5 Screens)

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
16	ArtisanOrdersScreen	features/artisan_orders/presentation/artisan_orders_screen.dart	Orders Loaded	[ArtisanOrdersScreen] Orders loaded: count=\$count	Audits order repository load performance.
16	ArtisanOrdersScreen	features/artisan_orders/presentation/artisan_orders_screen.dart	Order Item Click	[ArtisanOrdersScreen] Order tapped: orderNumber=\$num, status=\$status, customer=\$customer	Tracks order detail inspection.
17	ArtisanEarningsScreen	features/artisan_earnings/presentation/artisan_earnings_screen.dart	build render	[ArtisanEarningsScreen] Rendered. Available balance: ₹8,720, isHindi: \$isHindi	Audits balance presentation and localization.
17	ArtisanEarningsScreen	features/artisan_earnings/presentation/artisan_earnings_screen.dart	Weekly Row Click	[ArtisanEarningsScreen] Weekly summary tapped: week=\$week, amount=\$amount, orders=\$orders	Audits weekly payout analysis.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	initState	[ArtisanProfileScreen] Initialized	Validates profile mount.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Aadhaar Toggle	[ArtisanProfileScreen] Aadhaar toggle tapped. isRevealed: \$isRevealed	Audits security mask on sensitive artisan Aadhaar number.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Play Story Audio	[ArtisanProfileScreen] Story audio play toggle: isPlaying=\$isPlaying	Audits artisan bio audio narration engagement.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Edit Story Tap	[ArtisanProfileScreen] Edit story tapped → navigating to /artisan/story-input	Tracks story editing entry point.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	View Shop Catalog	[ArtisanProfileScreen] View shop catalog tapped for artisan: \$artisanId	Verifies navigation to artisan catalog.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Contact Artisan Dialog	[ArtisanProfileScreen] Contact artisan tapped → opening contact dialog	Logs direct contact inquiries.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Product Card Tap	[ArtisanProfileScreen] Product card tapped: id=\$productId	Profile craft showcase click telemetry.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Switch to Buyer Role	[ArtisanProfileScreen] Role switch to Buyer confirmed → navigating to /buyer/home	Captures mode-switching between artisan seller and buyer.
18	ArtisanProfileScreen	features/artisan_profile/presentation/artisan_profile_screen.dart	Sign Out Tap	[ArtisanProfileScreen] Sign out confirmed → navigating to /login	Audits session termination.
19	ArtisanStoryTextInputScreen	features/artisan_profile/presentation/artisan_story_text_screen.dart	initState	[ArtisanStoryTextInputScreen] Initialized	Verifies written story form mount.
19	ArtisanStoryTextInputScreen	features/artisan_profile/presentation/artisan_story_text_screen.dart	Form Submission	[ArtisanStoryTextInputScreen] Submitting story form: name="\$name", craft="\$craft", storyLength=\$len	Audits artisan bio updates and content lengths.
19	ArtisanStoryTextInputScreen	features/artisan_profile/presentation/artisan_story_text_screen.dart	Return to Profile	[ArtisanStoryTextInputScreen] Returned to profile from dialog	Confirms dialog dismissal and profile refresh.
20	VoiceOnboardingScreen	features/artisan_profile/presentation/voice_onboarding_screen.dart	initState	[VoiceOnboardingScreen] Initialized	Verifies voice bio onboarding mount.
20	VoiceOnboardingScreen	features/artisan_profile/presentation/voice_onboarding_screen.dart	Mic Hold Down	[VoiceOnboardingScreen] Mic pressed down → listening started	Audits push-to-talk mic initiation.
20	VoiceOnboardingScreen	features/artisan_profile/presentation/voice_onboarding_screen.dart	Mic Release	[VoiceOnboardingScreen] Mic released → listening stopped, voice recorded	Audits push-to-talk completion.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
20	VoiceOnboardingScreen	features/artisan_profile/presentation/voice_onboarding_screen.dart	Prefer Typing Link	[VoiceOnboardingScreen] Prefer typing pressed → navigating to /artisan/story-input	Tracks fallback from voice to standard typing.

Group 4: Fairs & Community (2 Screens)

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	initState	[ArtisanFairsScreen] Initialized. Current tab index: \$index	Tracks fair directory entry.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Tab Selection	[ArtisanFairsScreen] Tab selected: index=\$index, label="\$label"	Audits filtering across Upcoming, Applied, and Subsidized tabs.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Search Query	[ArtisanFairsScreen] Search query changed: "\$val"	Live search telemetry in fairs.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Voice Search Modal	[ArtisanFairsScreen] Opening voice search modal	Voice search engagement metric.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Bookmark Fair	[ArtisanFairsScreen] Bookmark toggled for fair: id=\$id, newState=\$state	Audits fair saving behavior.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	View Details Click	[ArtisanFairsScreen] Navigating to fair details: /fair/\$fairId	Tracks transition to fair details.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Apply for Stall	[ArtisanFairsScreen] Stall application submitted for fair: id=\$id, title=\$title	Audits artisan exhibition participation.
21	ArtisanFairsScreen	features/artisan_fairs/presentation/artisan_fairs_screen.dart	Support Helpline	[ArtisanFairsScreen] Calling helpline 1800-200-8899	Tracks support phone dialer trigger.
22	FairDetailScreen	features/artisan_fairs/presentation/fair_detail_screen.dart	initState	[FairDetailScreen] Initialized for fairId: \$fairId	Validates fair detail screen initialization.
22	FairDetailScreen	features/artisan_fairs/presentation/fair_detail_screen.dart	View Artisan Shop	[FairDetailScreen] View Shop clicked for artisan: \$artisanId	Verifies Task 1 navigation fix from fair to shop catalog.
22	FairDetailScreen	features/artisan_fairs/presentation/fair_detail_screen.dart	Featured Craft Tap	[FairDetailScreen] Featured product card tapped: id=\$id, name="\$name"	Tracks craft discovery within fair.
22	FairDetailScreen	features/artisan_fairs/presentation/fair_detail_screen.dart	Stall Apply Dialog	[FairDetailScreen] Stall application completed for fair: \$fairId	Verifies exhibition registration flow.

Group 5: Marketplace & Buyer Flow (5 Screens)

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	initState	[MarketplaceHomeScreen] Initialized	Tracks buyer marketplace launch.
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	Cart Icon Tap	[MarketplaceHomeScreen] Cart icon tapped (count=\$count) → navigating to /customer/cart	Cart inspection funnel.
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	Pull to Refresh	[MarketplaceHomeScreen] Pull-to-refresh triggered	Audits feed refresh frequency.
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	B2B Wholesale Banner	[MarketplaceHomeScreen] B2B wholesale tapped → navigating to /customer/b2b/bulk-request	B2B quotation lead tracking.
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	Cluster Filter Chip	[MarketplaceHomeScreen] Cluster selected: "\$cluster"	Geographical cluster filter telemetry (e.g. Gorakhpur, Maheshwar).
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	Product Card Tap	[MarketplaceHomeScreen] Product card tapped: id=\$id, name="\$name"	Product page view conversion tracking.
23	MarketplaceHomeScreen	features/marketplace_home/presentation/marketplace_home_screen.dart	Quick Add to Cart	[MarketplaceHomeScreen] Quick add to cart: id=\$id, price=\$price	Audits quick-buy interactions.
24	ProductDetailScreen	features/marketplace_home/presentation/product_detail_screen.dart	build render	[ProductDetailScreen] Rendered. productId=\$productId, isBuyer=\$isBuyer	Verifies contextual buyer vs artisan rendering.
24	ProductDetailScreen	features/marketplace_home/presentation/product_detail_screen.dart	Add to Cart Tap	[ProductDetailScreen] Add to cart pressed: productId=\$productId, price=\$price	Tracks cart conversion (exclusive to Buyer flow).
24	ProductDetailScreen	features/marketplace_home/presentation/product_detail_screen.dart	Artisan Edit Listing	[ProductDetailScreen] Artisan edit listing tapped → navigating to pricing	Verifies artisan seller quick-edit capability.
25	CartScreen	features/cart_checkout/presentation/cart_screen.dart	build render	[CartScreen] Rendered. cartItemsCount=\$count, total=₹ \$total	Cart contents and billing breakdown audit.
25	CartScreen	features/cart_checkout/presentation/cart_screen.dart	Quantity +/-	[CartScreen] Quantity increased / decreased for product: \$productId	Tracks cart quantity modifications.
25	CartScreen	features/cart_checkout/presentation/cart_screen.dart	Empty Cart CTA	[CartScreen] Empty cart button tapped → navigating to /customer/home	Tracks bounce recovery from empty cart.

#	Screen Name	File Path	Event Trigger / Lifecycle	Log Message Pattern	Diagnostic / Study Purpose
25	CartScreen	features/cart_checkout/presentation/cart_screen.dart	Checkout Button	[CartScreen] Checkout confirmed for total: ₹\$total	Audits purchase checkout completion.
26	CustomerAccountScreen	features/marketplace_home/presentation/customer_account_screen.dart	build render	[CustomerAccountScreen] Rendered for user: \$name	Buyer account overview audit.
26	CustomerAccountScreen	features/marketplace_home/presentation/customer_account_screen.dart	Switch to Artisan Track	[CustomerAccountScreen] Role switch to Artisan → navigating to /artisan/home	Tracks persona transition to Artisan studio.
26	CustomerAccountScreen	features/marketplace_home/presentation/customer_account_screen.dart	Sign Out Button	[CustomerAccountScreen] Sign out confirmed → navigating to /login	Audits buyer sign out.
27	B2bBulkRequestScreen	features/b2b_bulk_request/presentation/b2b_bulk_request_screen.dart	initState	[B2bBulkRequestScreen] Initialized	B2B RFQ form mount.
27	B2bBulkRequestScreen	features/b2b_bulk_request/presentation/b2b_bulk_request_screen.dart	Craft Category Select	[B2bBulkRequestScreen] Craft category changed: \$val	Tracks bulk craft categories in demand.
27	B2bBulkRequestScreen	features/b2b_bulk_request/presentation/b2b_bulk_request_screen.dart	Submit Quotation	[B2bBulkRequestScreen] Quotation request submitted: company="\$company", craft="\$craft", qty="\$qty"	Logs B2B quotation submission inputs.
27	B2bBulkRequestScreen	features/b2b_bulk_request/presentation/b2b_bulk_request_screen.dart	Dialog OK Tap	[B2bBulkRequestScreen] Request dialog OK pressed → popping screen	Confirms quotation dialog acknowledgment.

3. How to Use for Study and Debugging

1. Grepping by Tag:

Every log statement starts with [`<ClassName>`]. When debugging a specific screen (e.g., `ArtisanCameraScreen`), use:

```
flutter logs | grep "\[ArtisanCameraScreen\]"
```

2. Monitoring State Transitions:

In multi-step wizards like **Add Item Wizard**:

[`ArtisanCameraScreen`] -> [`ImageCropperScreen`] -> [`PhotoReviewScreen`] -> [`VoiceDescribeScreen`] -> [`AiReviewScreen`] -> [`PricingScreen`]

You can trace an entire craft creation lifecycle from photo capture to price publication in a single log trace.

3. Role Context Verification:

Screens that adapt dynamically between Artisan and Buyer (`ProductDetailScreen`, `ShopCatalogScreen`, `ArtisanFairsScreen`) print `isBuyer=true/false` so you can verify access rules and UI permissions in real-time.